

CO3-AT1 Case Study

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Case Study 1 – Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Sol:

Data:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median, Mode

Mean

$$\frac{4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30}{9} = \frac{82}{9} = 9.11$$

Mean = ₹9.11 lakhs

Median

There are 9 values.

Middle (5th) value = 7

Median = ₹7 lakhs

Mode

Most repeated values are **5 and 8**

Mode = ₹5 lakhs and ₹8 lakhs

2. Range

$$30 - 4 = 26$$

Range = ₹26 lakhs

3. Variance and Standard Deviation

Variance

$$\frac{\sum (x - \bar{x})^2}{9} = 57.88$$

Standard deviation

$$\sqrt{57.88} = 7.61$$

Variance = 57.88

Standard Deviation = 7.61

4. Outlier

The salary **₹30 lakhs** is much larger than the others.

Outlier = ₹30 lakhs

5. Best Measure of Central Tendency

Median is best because the outlier (₹30 lakhs) increases the mean significantly, while the median better represents the typical employee salary.

Case Study 2 – Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

Sol:

Data:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Mean and Median

Mean

$$\frac{1505}{10} = 150.5$$

Mean = 150.5 ms

Median

$$\frac{132 + 135}{2} = 133.5$$

Median = 133.5 ms

2. Range

$$300 - 120 = 180$$

Range = 180 ms

3. Variance and Standard Deviation

Response Time (ms)	$((x - \bar{x})^2)$
120	930.25
125	650.25
128	506.25
130	420.25
132	342.25
135	240.25
140	110.25
145	30.25
150	0.25
300	22350.25

Variance

2590.25

Standard deviation

50.89

Sum of squared deviations = **25580.50**

Variance

$$\frac{25580.50}{10} = 2558.05$$

Variance = 2558.05 ms²

Standard Deviation

$$\sqrt{2558.05} \approx 50.58$$

Standard Deviation $\approx$ 50.58 ms

4. Skewness

Since one very high value (300) exists,

The data is right-skewed (positively skewed)

5. Why Median?

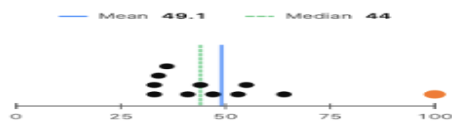
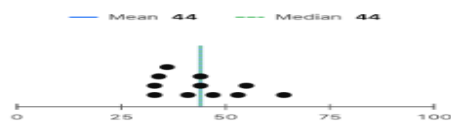
Median is not affected much by the extreme value (300 ms), so it better represents the normal website response time.

Case Study 3 – Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the IQR.
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.



Sol:

Data:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Q1, Q2 and Q3

Lower

42, 45, 48, 50, 52, 52

half:

$$Q_1 = \frac{48 + 50}{2} = 49$$

Median

$$Q_2 = \frac{52 + 55}{2} = 53.5$$

Upper half

55, 58, 60, 62, 65, 95

$$Q_3 = \frac{60 + 62}{2} = 61$$

Q1 = 49

Q2 = 53.5

Q3 = 61

2. IQR

$$61 - 49 = 12$$

IQR = 12

3. Outlier (1.5 × IQR Rule)

$$1.5 \times 12 = 18$$

Lower limit

$$49 - 18 = 31$$

Upper limit

$$61 + 18 = 79$$

Only **95** is greater than 79.

Outlier = 95

4. Mean and Median

Mean

$$\frac{684}{12} = 57$$

Mean = 57

Median

53.5

Median = 53.5

5. Effect of Outlier

The score **95** increases the mean, making it higher than the median.

Hence, the mean does not represent the average student performance as accurately as the median.

Case Study 4 – E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Sol:

Data:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Mean, Median and Mode

Mean

$$\frac{250}{10} = 25$$

Mean = 25

Median

$$\frac{24 + 25}{2} = 24.5$$

Median = 24.5

Mode

22

2. Variance and Standard Deviation

Orders (x)	$((x - \bar{x})^2)$
18	49
20	25
22	9
22	9
24	1
25	0
26	1
28	9
30	25
35	100

Sum of squared deviations = **228**

Variance

$$\frac{228}{10} = 22.8$$

Variance = 22.8

Standard Deviation

$$\sqrt{22.8} \approx 4.77$$

Standard Deviation ≈ 4.77 orders

3. Calculate the Coefficient of Variation (CV)

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$
$$= \frac{4.77}{25} \times 100 = 19.08\%$$

Coefficient of Variation $\approx 19.08\%$

4. What does Standard Deviation Tell?

The standard deviation shows how much the daily orders vary from the average number of orders.

5. Variability

Since **CV $\approx 18\%$** , the order data is **moderately variable**.

Case Study 5 – Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the Mean Error

Model A

$$\frac{2 + 3 + 3 + 4 + 4 + 5 + 5 + 6}{8} = \frac{32}{8} = 4$$

Model B

$$\frac{1 + 1 + 2 + 4 + 5 + 7 + 8 + 9}{8} = \frac{37}{8} = 4.625$$

Answer:

- Model A Mean = 4
- Model B Mean = **4.625**

2. Calculate the Variance (Population)

Model A

Mean = 4

Error	$(x - 4)^2$
2	4
3	1
3	1
4	0
4	0
5	1
5	1
6	4

Sum = **12**

$$\text{Variance} = \frac{12}{8} = 1.5$$

Variance = 1.5

Model B

Mean = 4.625

Error	$(x - 4.625)^2$
1	13.14
1	13.14
2	6.89
4	0.39
5	0.14
7	5.64
8	11.39
9	19.14

Sum $\approx$ **69.88**

$$\text{Variance} = \frac{69.88}{8} \approx 8.73$$

Variance ≈ 8.73

3. Calculate the Standard Deviation

$$\sqrt{1.5} = 1.22$$

$$\sqrt{9.23} = 3.04$$

Answer:

- Model A Standard Deviation = **1.22**

- Model B Standard Deviation = **3.04**

4. Which Model Has More Consistent Predictions?

Model A has the smaller standard deviation (1.22), so its prediction errors are closer to the mean.

Answer: Model A is more consistent.

5. Can the Model with the Lower Mean Error Be Considered Better?

No. A lower mean error alone does not mean a model is better. We should also consider **variance, standard deviation, consistency, and overall prediction performance**. A model with lower variability generally provides more reliable predictions.